



SRv6 in SONiC

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SRv6 in SONiC (Mature, Deployed, Rich Ecosystem)

- Mature:
 - [AI Backend: SDN Provisioned SRv6 Fabric](#)
 - L3VPN: IPv4/IPv6 – Static & BGP (RFC9252)
 - SRv6 GRT: IPv4/IPv6 – Static & BGP (RFC9252)
 - SRv6 L3 EVPN: IPv4/IPv6 (RFC9252)
 - Underlay Traffic Engineering: Static & IS-IS (RFC9352)
 - Static steering of IPv4/IPv6 traffic over SRv6 uSID
 - SRv6 SID Manager – Interoperability with IOS-XR/Other Vendors
 - BGP-LS: IS-IS and eBGP fabrics
 - 680+ PR and 2560+ commits across SAI/SONiC/FRR.
- Deployed in production:
 - Microsoft Fairwater DC – OCP 2025 [\[Link\]](#)
 - Alibaba eCore – OCP 2025 [\[Link\]](#)
- Rich Ecosystem:
 - Contributors: Cisco, Microsoft, Alibaba, Broadcom, Nvidia



Microsoft Fairwater DC
Guohan Lu @ OCP 2025



Alibaba eCore
Dennis Cai @ OCP 2025

SRv6 uSID in SONiC – Alibaba Deployment

- 680+ PR and 2560+ commits across SAI, SONiC, and FRR [mainline](#).
- 2021: [SRv6 uSID in SONiC collaboration started](#)
 - Very rich ecosystem (Alibaba, Cisco, Broadcom, Intel)
 - SRv6 uSID support in [SAI](#) and [SONiC](#)
- 2022: [Alibaba Routing Platform](#)
 - SONiC based and SRv6 uSID enabled
 - SRv6 uSID VPN support in [SONiC](#)
- 2023: [Alibaba SRv6 uSID deployment \(SONiC & IOS-XR\)](#)
 - Across all of China
 - SRv6 uSID VPN enhancement in [SAI](#) and [SONiC](#)
 - SONiC Routing WG ([Link](#)) and Alibaba & Cisco Blog ([Link](#))
- 2024: [SONiC PhoenixWing Initiative](#)
 - Endorsed by Alibaba, Cisco, Microsoft, Broadcom, Linux Foundation ([Link](#))
 - Enrich mainline SONiC with more SRv6 uSID features and testing

SRv6 uSID in SONiC – MSFT AI Backend

- SONiC 202505:
 - AI Backend - Static Programming of SRv6 uSID from SDN controller
 - A total of [122 PRs](#) have been merged across the [SAI, SONiC, and FRR mainline](#).
 - Cisco, Microsoft, Alibaba been the leading contributors.
 - SONiC Community blog: [LINK](#)

SONiC 202505: Powering AI Fabrics and Enterprise Networks with Precision and Insight

By SONiC | June 16, 2025 | No Comments

Author: Yanzhao Zhang

The SONiC 202505 release marks a major milestone in the evolution of open source networking, bringing powerful new capabilities tailored for AI backend fabrics, enterprise deployments, and next-generation observability.

This release introduces support for SRv6 (Segment Routing over IPv6) to enable source-routed AI backend networks—optimized for high-performance training and inference workloads. It also delivers several long-requested features for enterprise, campus, and edge data centers, making SONiC more versatile than ever.

Cisco – #1 Contributor of SRv6 in SONiC

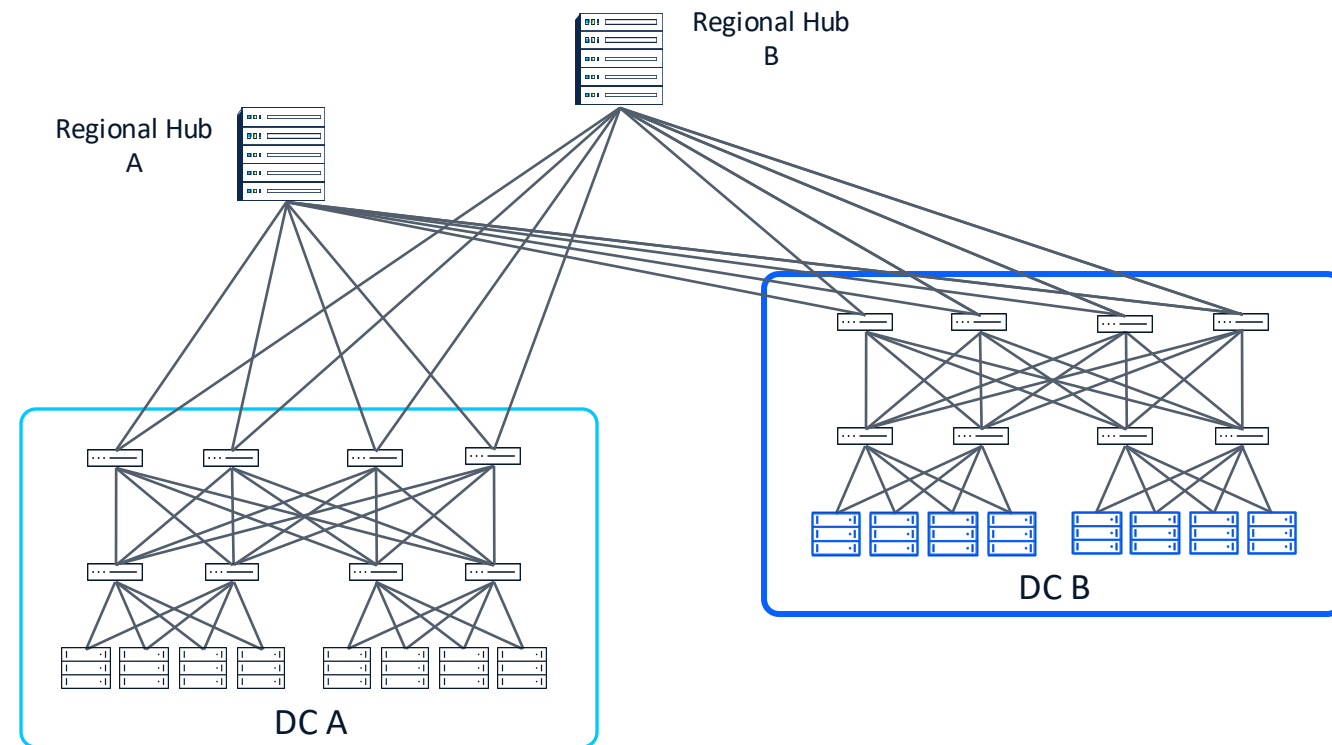
- 284 PR, 1204 commits across SAI/SONiC/FRR
- Cisco has the maintainer role for SRv6 uSID in FRR
 - SRv6 Static, BGP, SID Manager, IS-IS, ..
- Any SRv6 uSID PR in SONiC - reviewed/approved by Cisco



Evolving SRv6 in SONiC for AI DCI/WAN

DCI/WAN

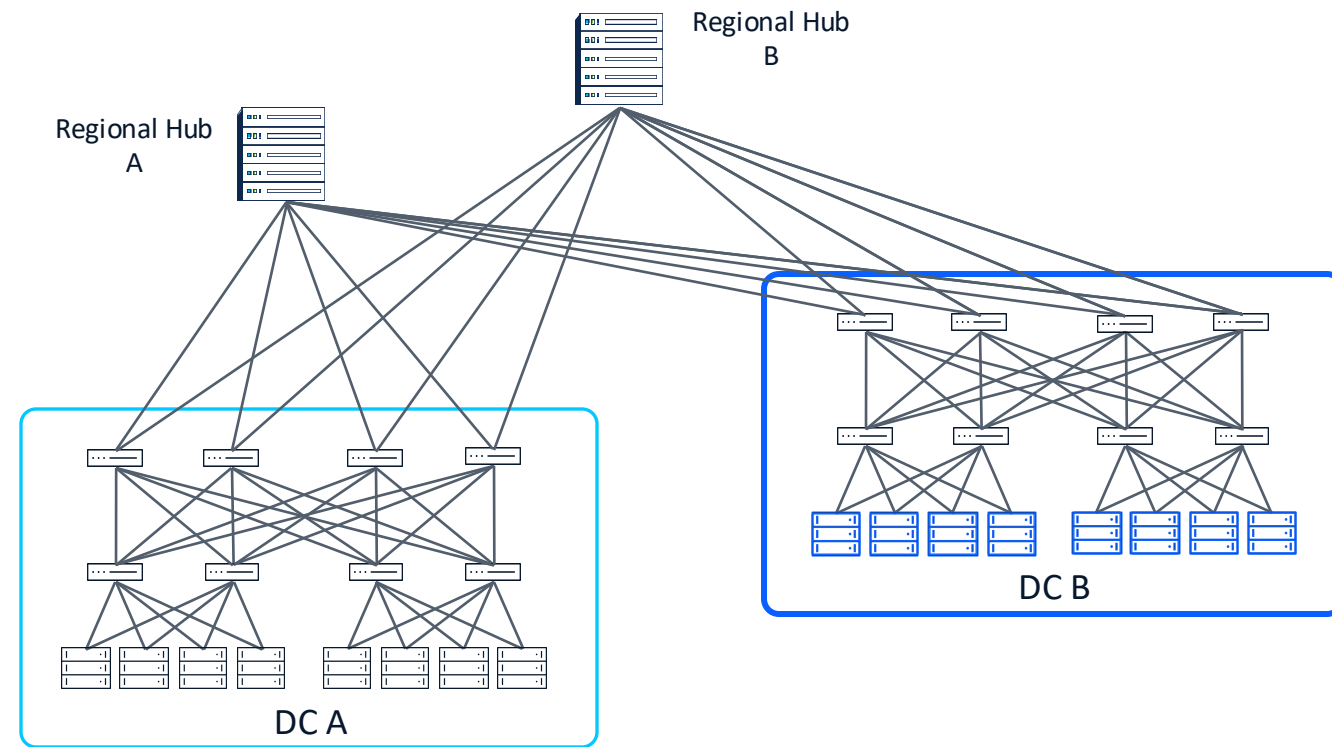
- Power limitations
- Workload spans clusters scattered over a regional network
- Traffic Engineering is MUST
 - Topology is less symmetric
 - Capacity is scarce
 - Latency matters
- SRv6 TE - Perfect fit
 - IP Nature of DCI
 - SONiC expansion in DCI
 - SRv6 rich/mature across DC HW
 - End-to-End - DC to DC through DCI



Clarence Filsfils, Guohan Lu, OCP EMEA 2026 [Link]

Topology is Essential

- TE path computation
 - Low latency, disjointness, BW
- IP Fast Reroute
 - LFA, Ti-LFA, uloop avoidance

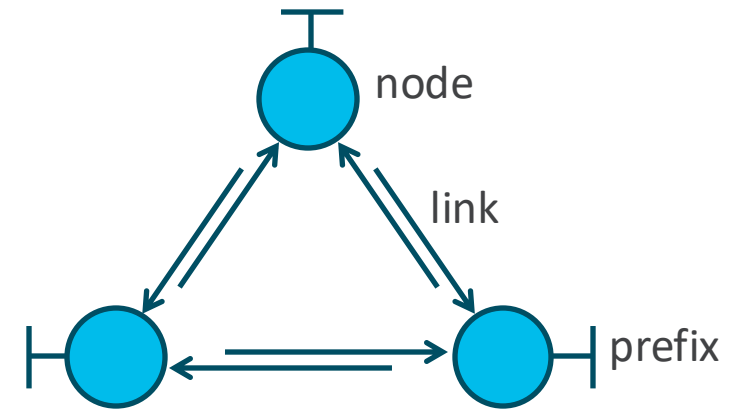


Topology view in SONiC

- Problem:
 - Existing approaches: *non-standard, fragmented, static, and difficult to scale across heterogeneous environments*
 - SONiC eBGP Fabric: each node has a visibility only of its direct neighbors
- Goal:
 - Standard, Open, Multi-vendor, and Scalable way.
 - Unified solution for IGP and eBGP Fabric

BGP-LS: the right tool

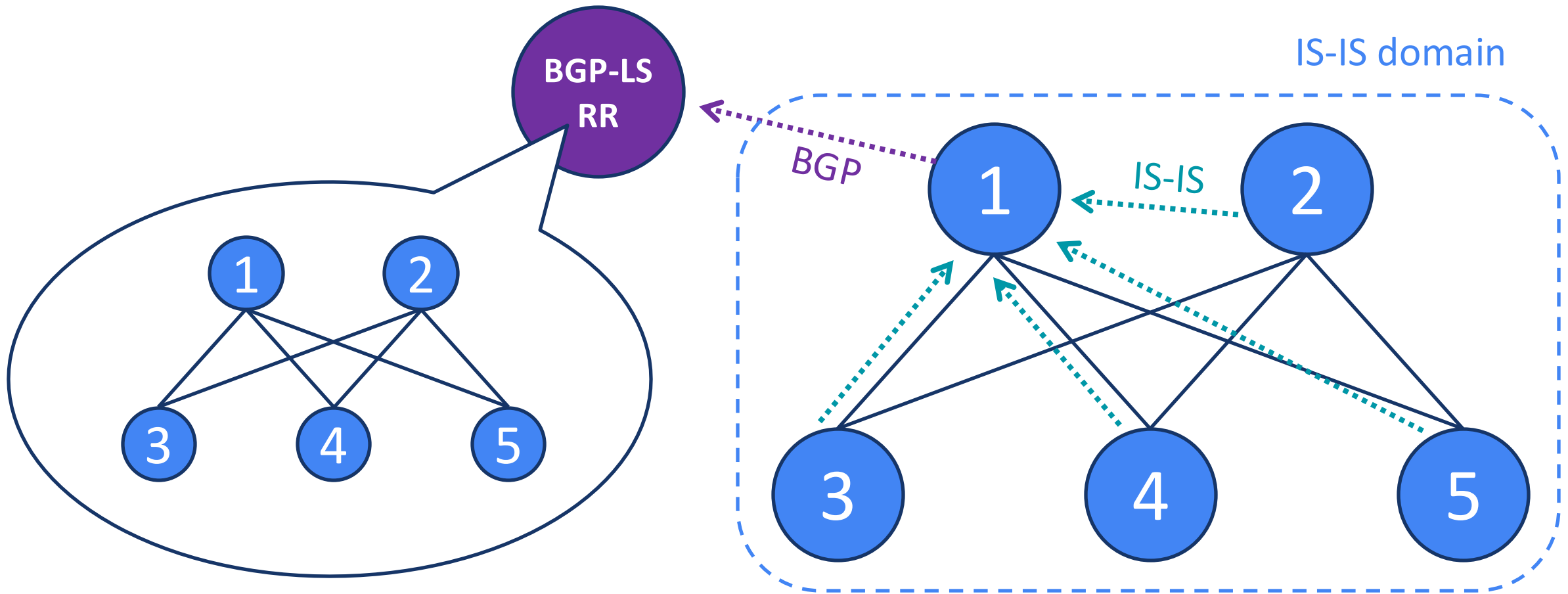
- Initially only for LS-IGPs...today much more...
- IETF RFC 9552 – a BGP extension to:
 - **Collect** topology info from the network protocol (IGP, BGP)
 - **Normalize** protocol information into a single topology model
 - **Distribute** it to any app or process that requires the topology
- BGP-LS topology model, three base object types:
 - **Node** – router, switch, protocol instance
 - **Link** – directed link anchored to two nodes
 - **Prefix** – anchored to a node
- Different BGP NLRI type per object-type
- More object/NLRI types exist (SID, SR Policy, ...)



Why BGP-LS?

- **Standard-based** – IETF RFC 9552
- **Multi-vendor** – Supported by majority of vendors
- **Open** – Supported by open-source stacks (e.g., FRR)
- **Scalable** – leverage mature BGP infrastructure (RRs, policies, filtering, update-groups)
- **Multi-domain** – consolidates topology across IGPs, BGP ASes
- **Rich** – carries topology graph, node and link attributes, SIDs, ...
- **Application-friendly** – one session to an RR yields the full topology graph
- **Control-plane only** – no data-plane impact

BGP-LS in IGP network (IS-IS) – deployment model



BGP-LS in IGP (IS-IS) – FRR support

- Available in FRR mainline - <https://github.com/FRRouting/frr/pull/20470>
- Supported since FRR 10.7.0 – released July 15, 2026

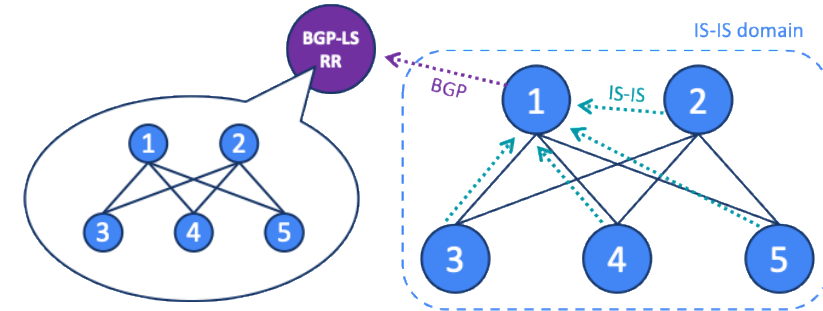
RR

```
router bgp 65000
neighbor 10.0.1.1 remote-as 65000
!
address-family link-state link-state
neighbor 10.0.1.1 activate
exit-address-family
```

R1

```
router bgp 65001
neighbor 10.0.0.1 remote-as 65000
!
address-family link-state link-state
neighbor 10.0.0.1 activate
exit-address-family

router isis SONIC
net 49.0001.0000.0000.0001.00
distribute link-state
!
```



BGP-LS Topology – NLRI's & Attributes

Node NLRI:

Protocol-ID:
Identifier:
Local Node:
AS:
BGP-LS ID:
Router-ID:
Area-ID:

Attributes:

Name:
Flags:
SRGB:
SRLB:
MSD:
IPv6-ID:

Link NLRI:

Protocol-ID:
Identifier:
Local Node:
AS:
BGP-LS ID:
Router-ID:
Area-ID:

Remote Node:

AS:
BGP-LS ID:
Router-ID:
Area-ID:

Link Descriptor:

Link Local ID:
Link Remote ID:

IPv4 Interface Address:
IPv4 Neighbor Address:
IPv6 Interface Address:
IPv6 Neighbor Address:

Attributes:

Metric:
TE Metric:
SRLG:
IGP Flags:
Delay:
.....

Prefix NLRI:

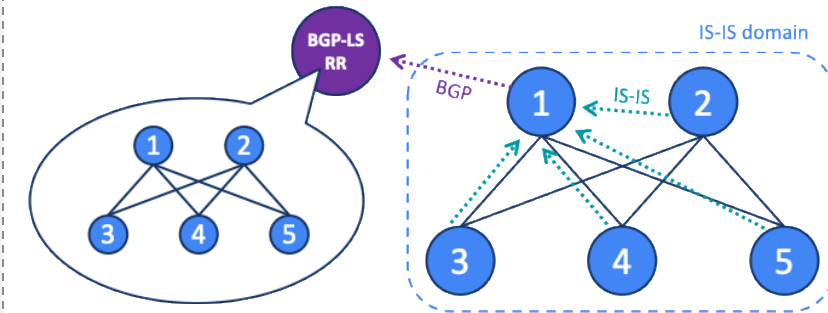
Protocol-ID:
Identifier:

Local Node:
AS:
BGP-LS ID:
Router-ID:
Area-ID:

Prefix Descriptor:
IP Reachability Info:
Prefix Length:
OSPF Route Type:
IGP Flags:

Attributes:

Metric:
Prefix Metric:
OSPF Forwarding Address:
Route Tag:
Extended Route Tag:
Prefix SID:
SR Algorithm:
SR Local Block:
SR Mapping Server:
IGP Flags:
Route Type:



BGP-LS – FRR Show CLI

- **Display all NLRIs**

```
show bgp link-state link-state
```

- **Show a specific Node NLRI**

```
show bgp link-state link-state [NODE NLRI]
```

- **Show a specific Link NLRI**

```
show bgp link-state link-state [LINK NLRI]
```

- **Show a specific Prefix NLRI**

```
show bgp link-state link-state [Prefix NLRI]
```

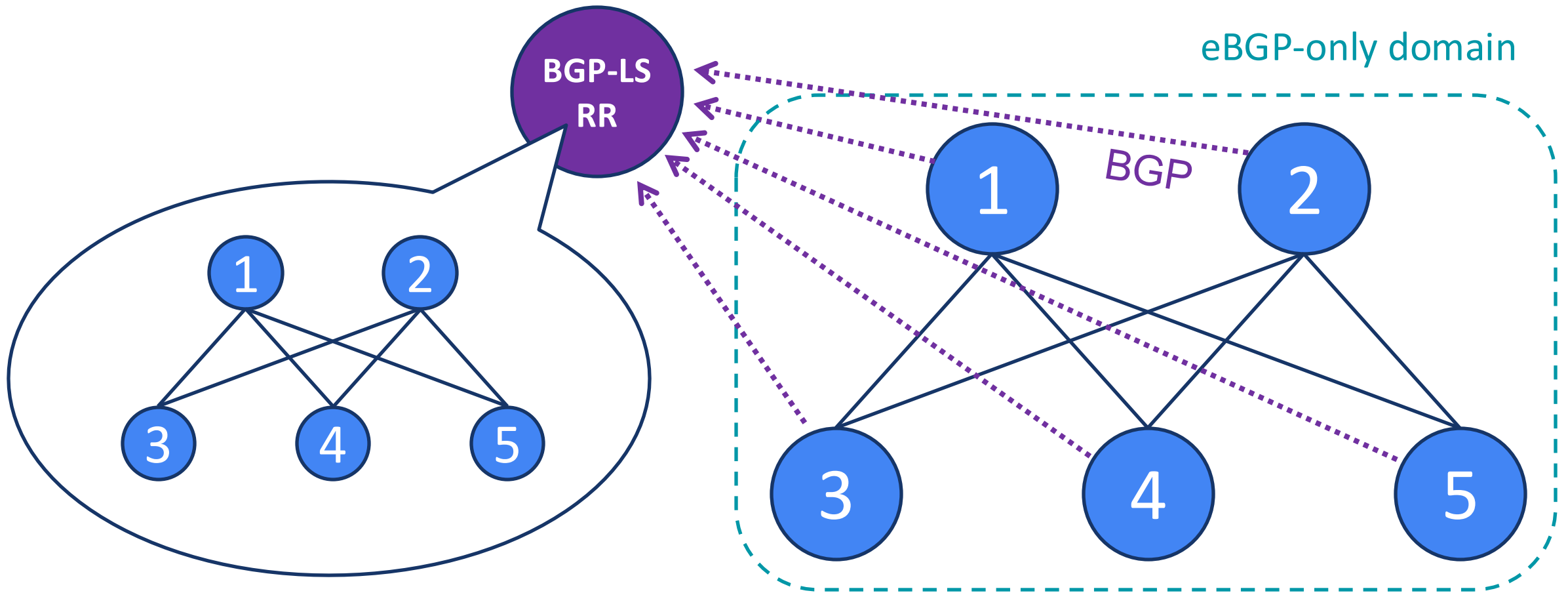
SONiC eBGP Fabric

- eBGP is the norm in both DC and DCI
- Data Center Fabrics
 - eBGP-only Clos designs
 - No IGP required — BGP handles both underlay and overlay reachability
- Data Center Interconnect
 - eBGP extends naturally beyond the fabric boundary, across DC and DCI domains
 - Simplified operations: one protocol, one toolchain

BGP-LS for eBGP

- No need for IGP
 - BGP-LS exports topology of eBGP fabrics
- Same BGP-LS topology model objects used in IGP
 - Node, Link, Prefix, SID, SR Policy,
- Standard
 - draft-ietf-idr-bgp-ls-bgp-only-fabric
 - specifies how to encode and advertise an eBGP-only network topology in BGP-LS
- Supported in FRR mainline
- Fully compatible with existing BGP-LS consumers
 - No changes to BGP-LS RR, Controller, or Apps

BGP-LS for eBGP – deployment model



BGP-LS for eBGP – FRR support

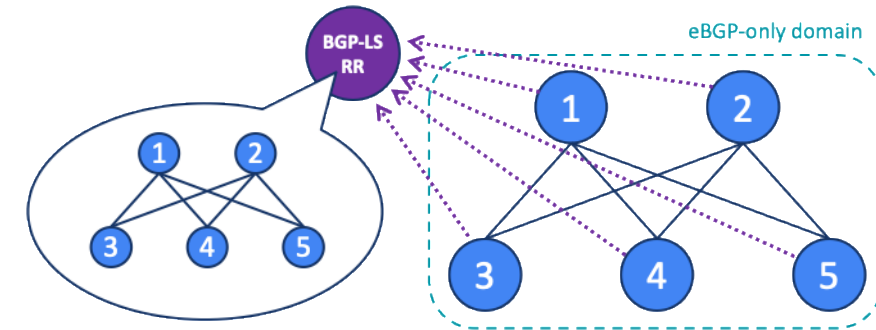
- Available in FRR mainline - <https://github.com/FRRouting/frr/pull/20726>
- Supported since FRR 10.7.0 – released July 15, 2026

RR

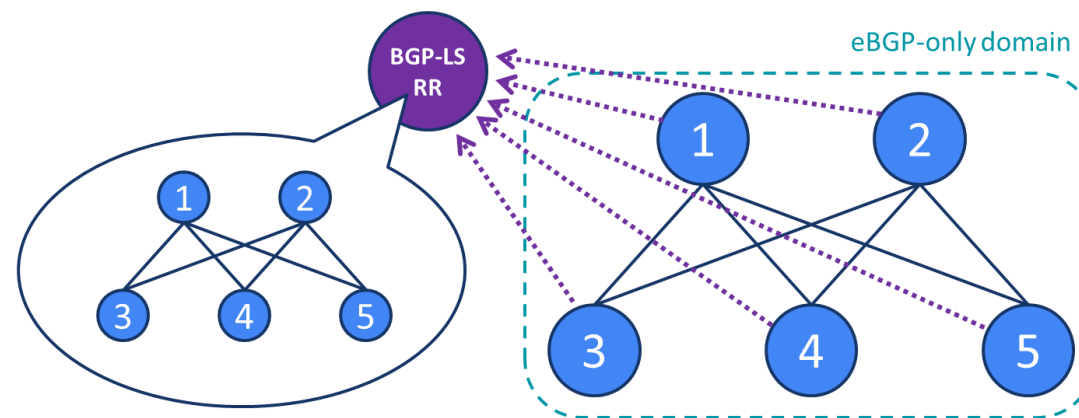
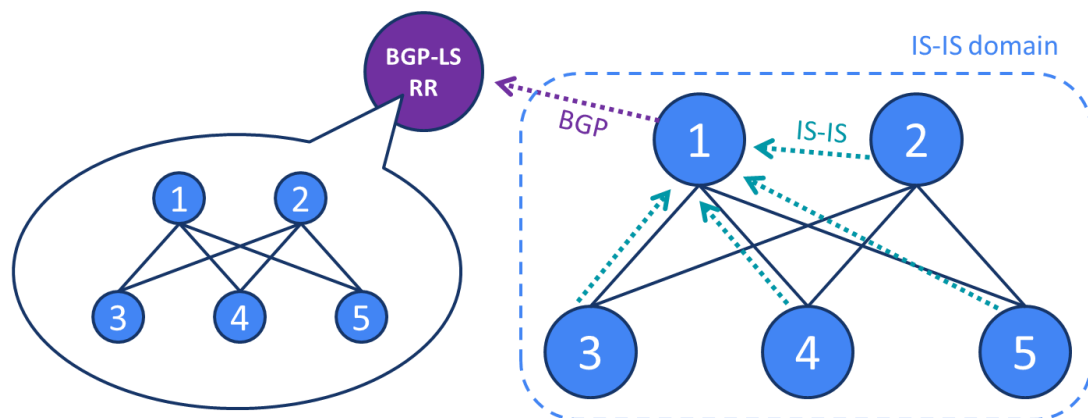
```
router bgp 65000
neighbor 10.0.1.1 remote-as 65000
neighbor 10.0.2.1 remote-as 65000
neighbor 10.0.3.1 remote-as 65000
neighbor 10.0.4.1 remote-as 65000
neighbor 10.0.5.1 remote-as 65000
!
address-family link-state link-state
neighbor 10.0.1.1 activate
neighbor 10.0.2.1 activate
neighbor 10.0.3.1 activate
neighbor 10.0.4.1 activate
neighbor 10.0.5.1 activate
exit-address-family
```

R1 – R5

```
router bgp 65000
neighbor 10.0.0.1 remote-as 65000
!
address-family link-state link-state
distribute bgp-fabric-link-state
neighbor 10.0.0.1 activate
exit-address-family
```



BGP-LS - Unified solution for topology view in IGP & BGP Fabrics



Node NLRI:

Protocol-ID:
Identifier:
Local Node:
AS:
BGP-LS ID:
Router-ID:
Area-ID:

Attributes:

Name:
Flags:
SRGB:
SRLB:
MSD:
IPv6-ID:

Link NLRI:

Protocol-ID:
Identifier:
Local Node:
AS:
BGP-LS ID:
Router-ID:
Area-ID:
Remote Node:

AS:
BGP-LS ID:
Router-ID:
Area-ID:
Link Descriptor:
Link Local ID:
Link Remote ID:
IPv4 Interface Address:
IPv4 Neighbor Address:
IPv6 Interface Address:
IPv6 Neighbor Address:

Attributes:

Metric:
TE Metric:
SRLG:
IGP Flags:
Delay:
.....

Prefix NLRI:

Protocol-ID:
Identifier:
Local Node:
AS:
BGP-LS ID:
Router-ID:
Area-ID:

Prefix Descriptor:
IP Reachability Info:
Prefix Length:
OSPF Route Type:
IGP Flags:

Attributes:

Metric:
Prefix Metric:
OSPF Forwarding Address:
Route Tag:
Extended Route Tag:
Prefix SID:
SR Algorithm:
SR Local Block:
SR Mapping Server:
IGP Flags:
Route Type:

Node NLRI:

Protocol-ID:
Identifier:
Local Node:
AS:
BGP-LS ID:
Router-ID:
Area-ID:

Attributes:

Name:
Flags:
SRGB:
SRLB:
MSD:
IPv6-ID:

Link NLRI:

Protocol-ID:
Identifier:
Local Node:
AS:
BGP-LS ID:
Router-ID:
Area-ID:
Remote Node:

AS:
BGP-LS ID:
Router-ID:
Area-ID:
Link Descriptor:
Link Local ID:
Link Remote ID:
IPv4 Interface Address:
IPv4 Neighbor Address:
IPv6 Interface Address:
IPv6 Neighbor Address:

Attributes:

Metric:
TE Metric:
SRLG:
IGP Flags:
Delay:
.....

Prefix NLRI:

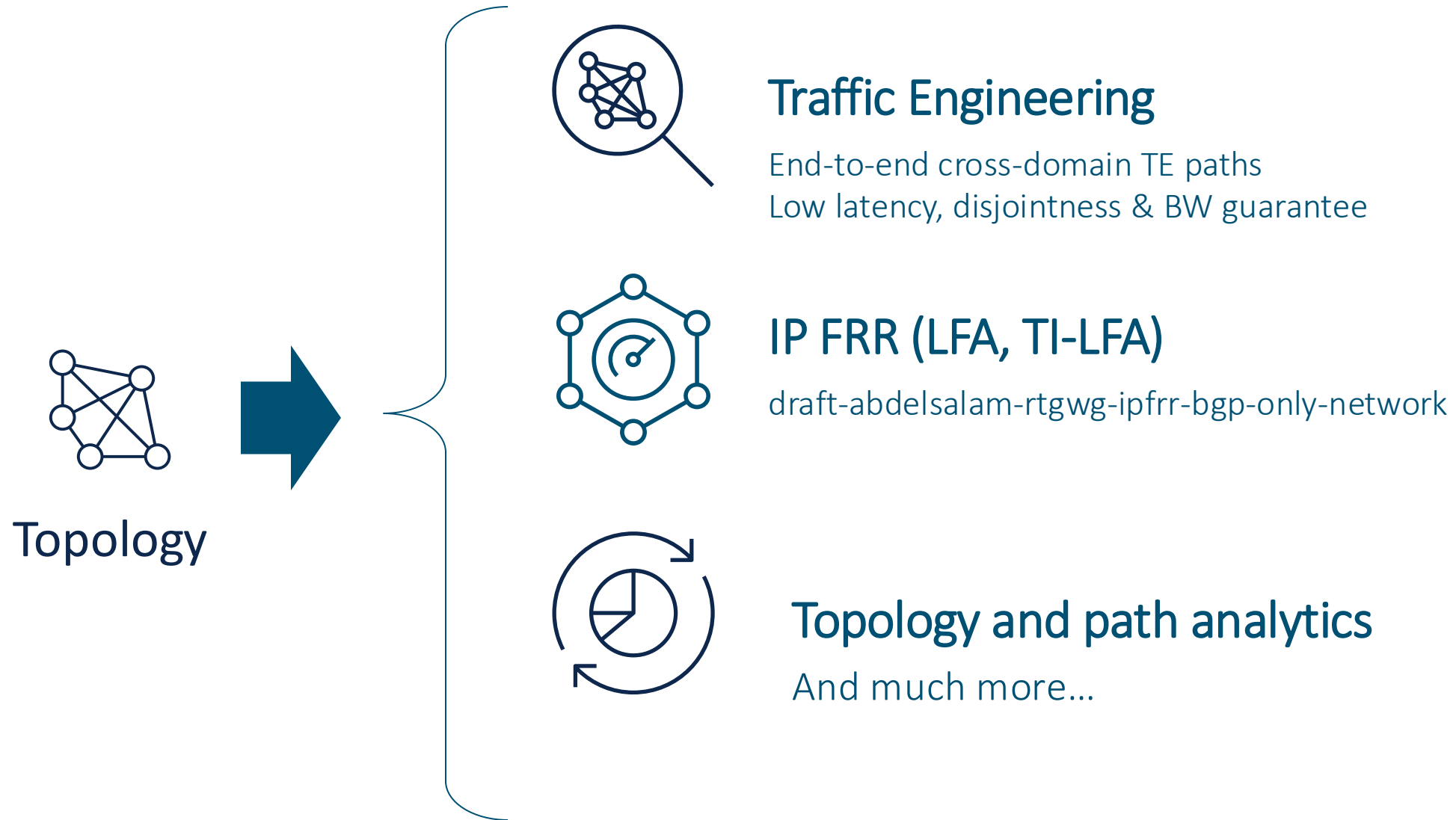
Protocol-ID:
Identifier:
Local Node:
AS:
BGP-LS ID:
Router-ID:
Area-ID:

Prefix Descriptor:
IP Reachability Info:
Prefix Length:
OSPF Route Type:
IGP Flags:

Attributes:

Metric:
Prefix Metric:
OSPF Forwarding Address:
Route Tag:
Extended Route Tag:
Prefix SID:
SR Algorithm:
SR Local Block:
SR Mapping Server:
IGP Flags:
Route Type:

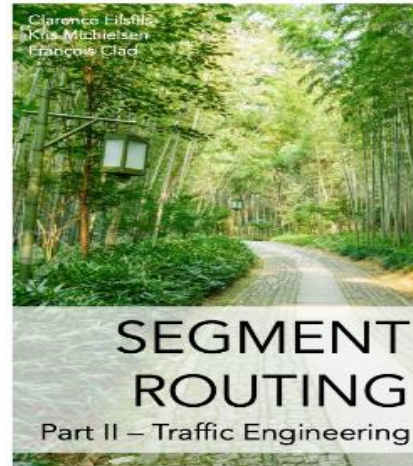
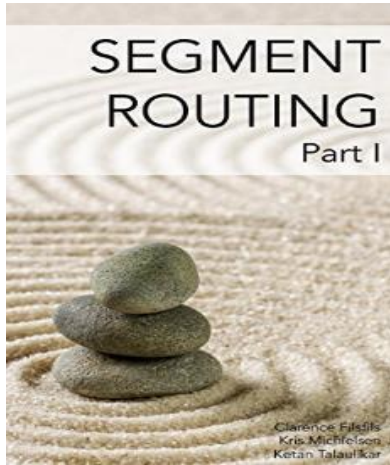
Foundational Building Block



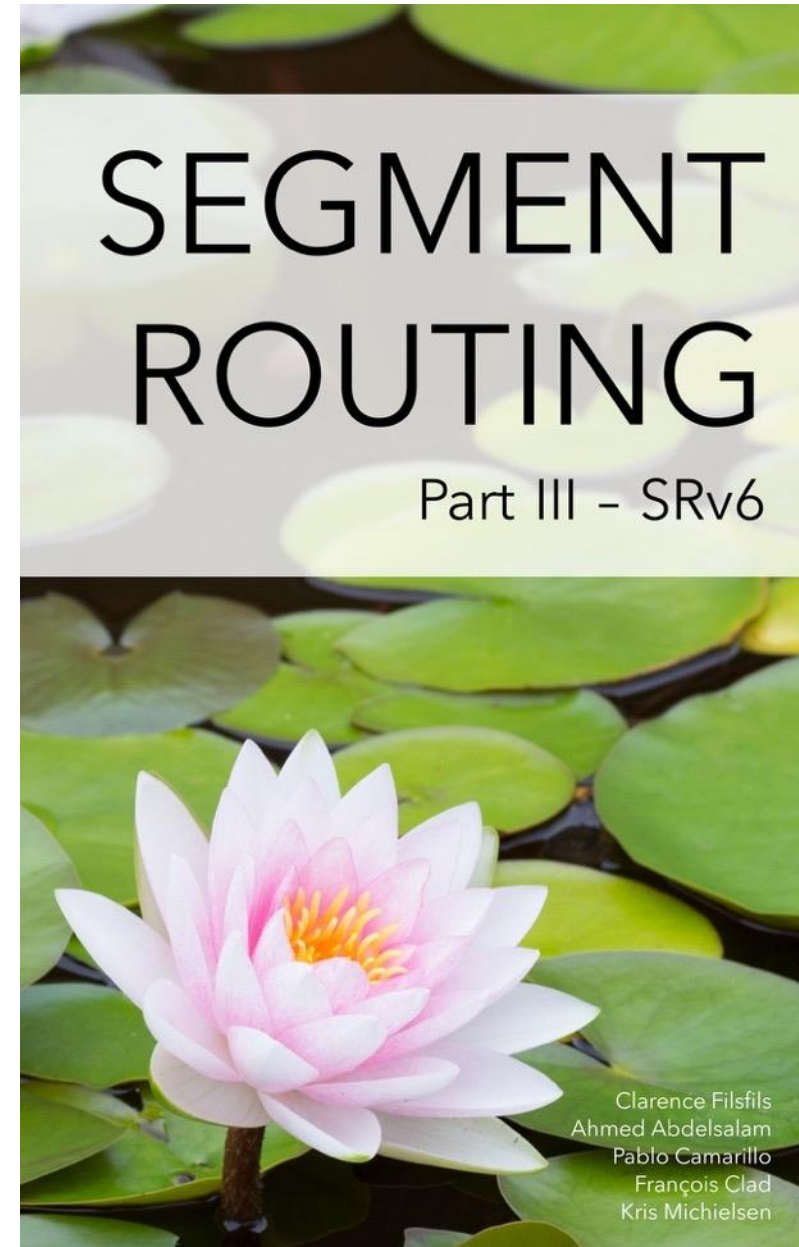
Conclusion

- SRv6 in SONiC:
 - Mature
 - Deployed
 - Rich Ecosystem
- Cisco Commitment and Leadership
 - #1 Contributor of SRv6 in SONiC
 - Cisco has the maintainer role for SRv6 in FRR
- Evolving SRv6 in SONiC for AI DCI/WAN
 - Topology view with BGP-LS for eBGP fabric
 - IP FRR (LFA, TI-LFA) for eBGP Fabric
 - Traffic Engineering and Analytics

Stay up-to-date



segment-routing.net



<https://www.amazon.com/dp/B0D6GWWRWH>

